
CE 569P: Structural Design Studio

Course Code: CE 569P

Course Name: Structural Design Studio

L-T-P-C: 0-0-6-3

Prerequisites: CE 351, CE 353P, CE 401

Intended for: UG/PG Elective

Mutual Exclusion: None

Approval: 61st BoA

Preamble

The Design Studio course is designed to bridge the gap between theoretical knowledge and practical application in the field of Structural Engineering. In today's design and construction industry, proficiency in both traditional design principles and modern computational tools is essential. This course provides students with hands-on experience in using advanced software packages for the analysis and design of both reinforced concrete (RC) and steel structures. Through a combination of computer programming, CAD-based drawing development, software packages, and comprehensive design documentation, students will acquire the necessary skills to tackle complex structural engineering challenges.

The course is meticulously structured to guide students through the entire design process, from initial conceptualization to the final production of detailed construction drawings and design reports. Emphasis is placed on the development of automated design tools and the application of relevant IS Standards, ensuring that students are well-equipped to meet the demands of the profession. By the end of the course, students will have enhanced their technical competencies and ability to integrate innovative solutions into real-world engineering projects.

On successful completion of the course, the student will be able to:

- **Develop Customized Programs:** Write and implement computer programs for analyzing trusses, beams, and frames.
- **Create Architectural and Structural Drawings:** Use CAD software to produce architectural and structural general arrangement (GA) drawings, including plans, sections, and elevations.
- **Perform Structural Analysis:** Set up and analyze numerical models for RC buildings and steel structures, considering earthquake and wind loads.
- **Design Structural Components:** Design RC and steel members using relevant IS Standards and develop automated Excel design sheets.
- **Produce Project Documentation:** Prepare detailed construction drawings, including reinforcement detailing, and compile comprehensive design reports.
- **Apply Theoretical Knowledge Practically:** Integrate theoretical concepts with practical applications in structural engineering.

Course Contents

The course content is split into several computer applications of different structural engineering analysis and design problems. Details are given below:

- **Development of a Computer Program for Analysis of 2D and 3D Truss** [6 Hours]
- **Development of a Computer Program for the Analysis of Beams and Frames** [6 Hours]
- **Review of Structural Design and Drawings of a Completed Project** [6 Hours]
- **Analysis and Design of Reinforced Concrete (RC) Multistorey Buildings in Software Packages:** [30 Hours]
 - a. Development of Architectural GA Drawings using CAD software - Include plan, sections, and elevation
 - b. Development of Structural GA Drawings using CAD software - Include layout of columns, beams, slabs, and foundations
 - c. Development of Numerical Model in Software Packages - Model Setup; Specification of Material and Section Properties; Loading and Boundary conditions; Load cases and Load combinations; Earthquake and Wind Analysis
 - d. Design of RC Members (Beams, Slabs, Columns, Shear Walls and Foundations) using relevant IS Standards, Development of Automated Excel Design Sheets
 - e. Development of Construction Drawings of the Buildings - Ductile Detailing, Reinforcement Drawings, Bill of Quantities
 - f. Development of Design Report with all Loads Calculations, Drawings, Design, and relevant Documentation.
- **Analysis and Design of Industrial Steel Structures in Software Packages:** [24 Hours]
 - a. Selection of Appropriate Configuration of the Industrial Steel Shed - Trusses or Moment Resisting Frames
 - b. Case-study Industrial Steel Shed - Development of Structural GA Drawings using Steel Sections in CAD software
 - c. Development of Numerical Model in Software Packages - Model Setup; Specification of Material and Section Properties; Loading and Boundary Conditions; Load cases and Load combinations; Connection Boundary Conditions; Earthquake and Wind Analysis
 - d. Design of Steel Members (Beams, Columns, Purlins, Rafter, Girts and Ties); Design of Moment and Shear Connection; Design of Base Plate; Development of Automated Excel Design Sheets
 - e. Development of Final GA Drawings showing Plan and Sections; Development of Connection Detailing Drawings, Fabrication Drawings
 - f. Development of Design Report with all Loads Calculations, Drawings, Design, and relevant Documentation.
- **Introduction to Analysis of Other Infrastructure Systems:** Short-Medium Span Bridges, Dams, Water Tanks, etc. [12 Hours]

Text Books

1. Shah, V. L., & Karve, S. R. (2005), *Illustrated design of reinforced concrete buildings*, Structures Publication.
2. Krishna Raju, N. (2011) *Structural Design and Drawing* (4th Edition), Universities Press



References

1. Weaver, W., and Gere, J. M. (2012), *Matrix analysis framed structures*, Springer science & business media.
2. Pillai, S.U. and Menon, D. (2009), *Reinforced concrete design*, Third Edition, McGraw Hill Education.
3. Subramanian, N. (2008). *Design of steel structures*, Oxford University Press.